

FACULTY OF ENGINEERING

INGB413: GA Assignment

By:

Mr SH Oberholzer

Student number: 37220020

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1 Background

Tangram puzzles are traditional Chinese puzzles that consist of seven geometric shapes that can be assembled in a variety of arrangements on an assembly board. In this assignment, tangram puzzles for a tangram manufacturing company, *Tangra Inc.*, are assembled using two different experimental procedures. These two experimental procedures are performed and evaluated to identify which procedure is best at identifying defective tangram puzzle pieces and which produces the most tangrams within the procedure time limit of 15 minutes. For this assignment, a specific arrangement of coloured triangles, a rhombus, and a square is assembled onto a square assembly board for shipment to customers. The principles and techniques of quality management and statistical process control are used to analyse and evaluate the quality parameters of the two procedures. Their effects on company profitability are also reported.

2 Data Collection and Analysis

Two processes were tested to evaluate the effectiveness of the detection of defective puzzle pieces and the assembly rate for tangram puzzles. A completed tangram puzzle is illustrated below in Figure 1.

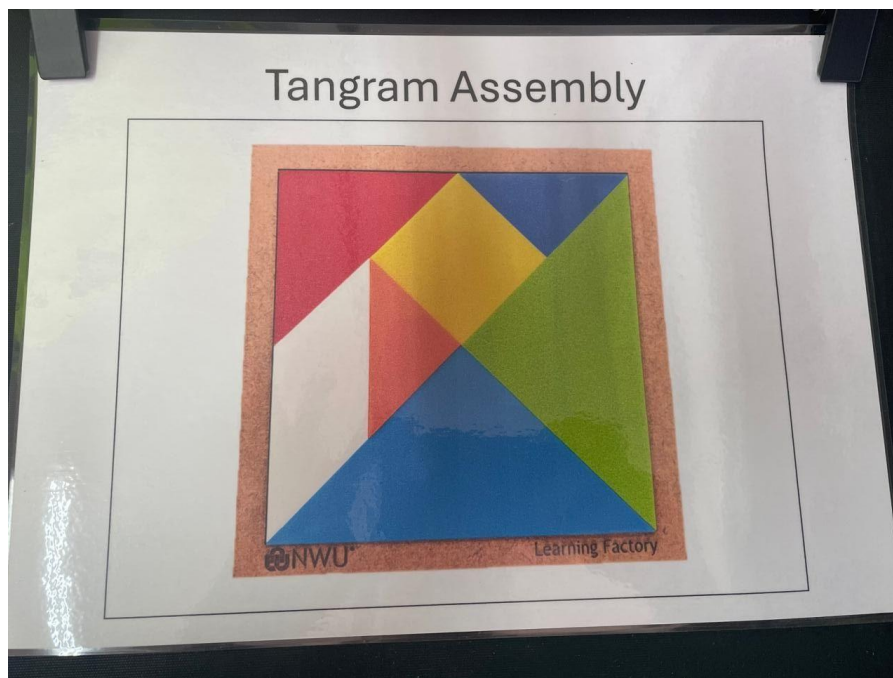


Figure 1: Picture of Assembled Tangram

Each process was tested on an individual assembly station with a stopwatch to ensure that the 15-minute time limit was adhered to. In addition, each assembly station had a size board with the correct shape outlines drawn on the board available to evaluate the pieces drawn for defects.

2.1 Process 1

2.1.1 Process 1: Procedure

The first process tested followed the following steps to complete a tangram assembly:

1. Receive raw material in bulk in a randomly organised bin.
2. Take random samples of five pieces grouped by colour and associated shape (i.e. five red triangles).
3. Place each piece on the size board and evaluate if the piece fits in the shape outline (non-defective) or not (defective).
 - 3.1 If a piece is defective, it is placed within a bin labelled "Defects".
 - 3.2 If a piece is non-defective, it is placed to the side until enough pieces are available for a complete assembly. Groups of non-defective pieces are stacked upon one another.
4. Random sampling is repeated until at least one piece of each kind is available for a complete assembly.
5. An assembly board is drawn from a bin, placed in the centre of the assembly station, and a tangram is assembled.
6. The assembled tangram is then moved to a designated complete assembly area to the right of the centre of the assembly station.

2.1.2 Process 1: Data Collection

As noted in the outlined procedure, the raw material was received in bulk within a bin. Each sample of 5 pieces was randomly done or by colour group when a shortage of a specific piece was noted. Each batch of five pieces was tested for defects and the amount of defects out of five was recorded. The associated colour group was also recorded for a drawn group of five pieces.

The total amount of defects and the number of completed tangrams were also recorded. From the recorded results, the proportions of defects for each sample were calculated and the proportion of overall defects was also calculated.

The number of defects per batch (d_i) for each constant size batch of $n = 5$ pieces has a range of discrete values of $d_i \in \{0, 1, 2, 3, 4, 5\}$. The colour groups tested and recorded, as seen in Figure 1, are Red (R), Yellow (Y), Orange (O), Light Blue (LB), Light Green (LG), Dark Blue (DB) and White (W).

The proportion of defects (p_i) are calculated from the collected data with:

$$p_i = \frac{d_i}{n} = \frac{d_i}{5} \quad (1)$$

The overall defect rate for process 1 is similarly calculated as a point estimate with:

$$\hat{p}_1 = \frac{\sum d_i}{n} \quad (2)$$

Table 1: Sample Data for Process 1

Sample Number	n	Number of Defects (d_i)	Colour	Proportion of Defects (p_i)
1	5	4	R	0,8
2	5	3	Y	0,6
3	5	2	O	0,4
4	5	0	LB	0
5	5	0	LG	0
6	5	2	DB	0,4
7	5	1	W	0,2
8	5	5	R	1
9	5	3	R	0,6
10	5	2	Y	0,4
11	5	2	R	0,4
12	5	3	O	0,6
13	5	3	DB	0,6
14	5	4	W	0,8
15	5	5	Y	1
16	5	3	Y	0,6
17	5	4	O	0,8
18	5	2	W	0,4

Table 2: Summary Of Process 1 Data Collected

Total Samples	18
Batch Size	5
Total Pieces	90
Total Defects	48
Total Completed Tangrams	5
Overall Defect Proportion ($\hat{p}_1$)	0,5333

From Table 2, process 1 produced a total of 5 completed tangrams within the 15-minute time limit with a defect proportion of approximately 53.33 %.

2.1.3 Process 1: Data Visualisation

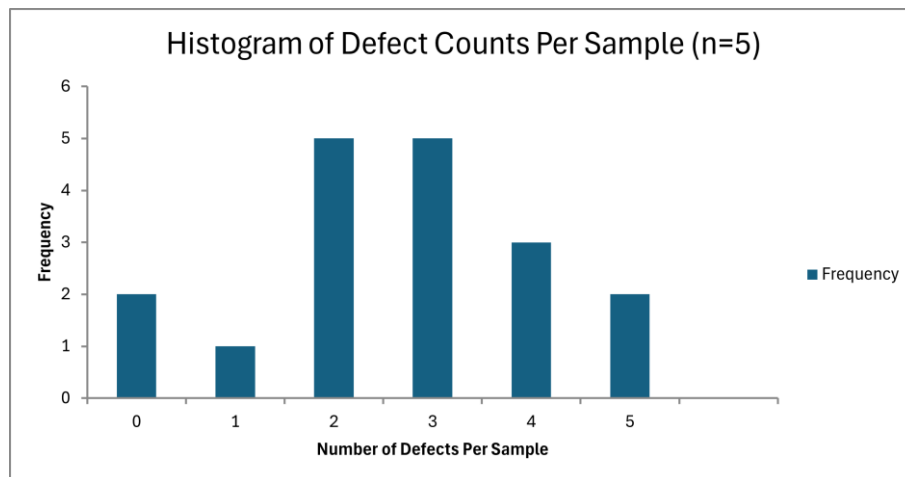


Figure 2: Histogram of Defect Counts per Sample (Process 1)

The number of defects recorded per sample from the recorded data in Table 1 is organised into bins in the histogram in Figure 2. The shape of the histogram is also roughly symmetrical, though it is slightly right-skewed.

The histogram of the defects counted out of each batch per sample indicates that the most frequently occurring defects were two and three pieces. Only two instances with zero defects were recorded, the least frequently occurring instance being one for one defect in a sample. Higher frequency defects also occur less frequently, with two instances of five defects per sample noted. Four defects per sample are observed to occur more frequently than five, but less frequently than two or three, with three recorded instances.

Overall, the histogram indicates that batches with very few defects (one or less) are less common. Defects are more likely to be between two and three per sample and less likely to be either four or five per sample in the first process. This implies that the frequency of defects ranges between 40% and 60%.

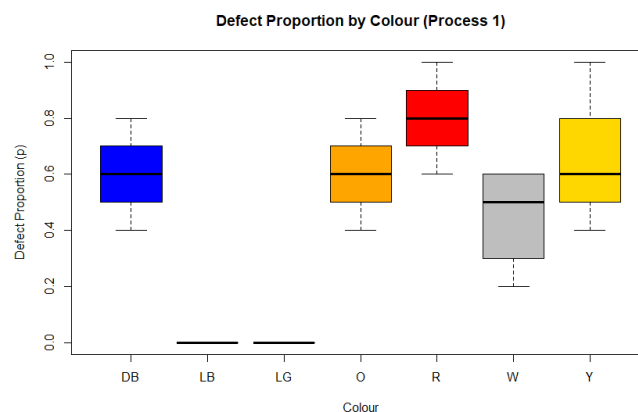


Figure 3: Box Plot of Defect Proportion by Colour (Process 1)

The box plot in Figure 3 shows the defect proportion per batch for each colour group. Of all colour groups, the light blue and light green colour groups indicated no defects drawn from the raw data. Excluding these two colour groups, the red colour group of pieces indicated the highest overall defect proportion with a median of approximately 0.8. The dark blue, orange, and yellow colour groups all have an approximately equal median defect proportion of 0.6. The white colour group, excluding light blue and light green, effectively has the lowest overall proportion of defects on average.

To confirm this observation from the box plot in Figure 3, the average defect rate per colour group (p_{avg}) is calculated and plotted on a bar graph in Figure 4.

Table 3: Average Defect Proportion per Colour Group

Colours	Samples	Samples per Colour	Defects	Average Proportion of Defects (p_{avg})
R	4	20	14	0,7
Y	4	20	13	0,65
O	3	15	9	0,6
LB	1	5	0	0
LG	1	5	0	0
DB	2	10	5	0,5
W	3	15	7	0,47

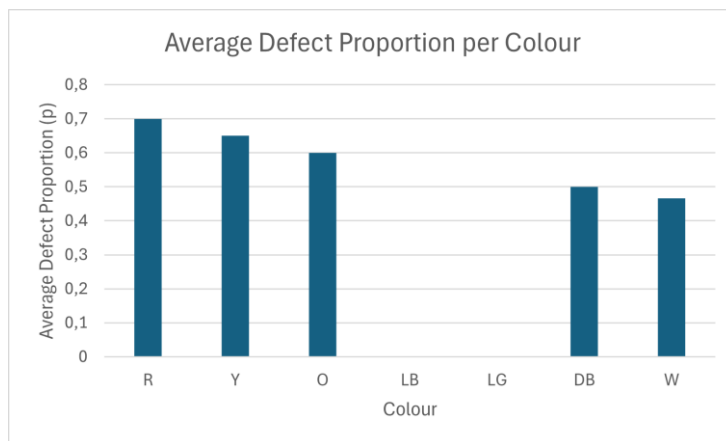


Figure 4: Bar Chart of Average Defect Proportion per Colour

Table 3 and Figure 4 confirm the initial observations from the box plot in Figure 3. The highest proportions of defects overall are (in order) red, yellow, orange, dark blue, and then white. Red pieces are identified as having the highest proportion of defects identified in this process, with yellow pieces having a very similar proportion to red pieces.

In general, more than half of all puzzle pieces from the bulk raw material received for Process 1 are defective. The highest defective pieces in out of the drawn pieces from the bulk lot are attributed to red and yellow pieces. Further investigation into the suppliers of these pieces and

the causes of these defects. The single drawn samples for light blue and light green pieces are not enough to conclusively state that they are the most high-quality pieces; however, in comparison to the other colour groups, they do yield the highest-quality pieces for assembly. Excluding these two colour groups indicates that the white pieces have the highest overall quality.

2.1.4 Process 1: Probability Distribution

For further analysis of process 1, the underlying distribution must be identified. For process 1 a fixed batch size per sample was taken of size $n = 5$ from a bin of mixed pieces. Each piece was tested and evaluated on a binary acceptance criteria; they either are non-defective and pass assessment, or they are defective and are rejected from the accepted lot of pieces for assembly.

In this process, we assume that each piece has the same probability of being defective (p_1), independent of other pieces within the same batch. Therefore, the recorded number of defects in each batch (d_i) is a measure of the number of defects in five independent trials with probability p_1 . With these assumptions, the procedure steps followed and the data gathered, the underlying probability distribution of process 1 can be concluded to be binomially distributed. With constant $n = 5$ and probability p_1 , the probability mass function is given by:

$$p(d_i = x) = \binom{5}{x} p_1^x (1 - p_1)^{5-x}, \quad x = 0, 1, 2, 3, 4, 5 \quad (3)$$

Using equation 1, the expected number of defects in each batch of 5 pieces drawn is given by:

$$E p_i = p_1 \quad (4)$$

$$\text{Var } p = \frac{p_1 (1 - p_1)}{5} \quad (5)$$

p_1 in this case is estimated using our estimate calculated with equation 2 at 0.53333, as indicated in Table 2.

2.1.5 Process 1: Normal Approximation to the Binomial Distribution

The binomial distribution can be approximated by a normal distribution for the purposes of constructing confidence intervals on $\hat{p}_1$ provided that two conditions are satisfied:

- $np \geq 5$ and
- $n(1 - p) \geq 5$

Our estimate for p , $\hat{p}_1$, is used to check with $n = 5$.

Table 4: Normal Approximation to the Binomial Distribution Results

n	5
$\hat{p}_1$	0,5333
np	2,6665
n(1-p)	2,3335

In both cases the evaluation metrics are less than five. The normal approximation to the binomial distribution in this case does not hold and the exact binomial estimate of variance will be used in further calculations.

2.1.6 Process 1: Statistical Control Analysis

To obtain an accurate estimate of $\hat{p}_1$ and be able to clearly compare the two processes used for this experiment, the stability of the process must be evaluated. Process 1 consists of samples with binary outcomes and a constant sample size. Additionally, an attribute was measured instead of continuous data, where the process also follows a binomial distribution. It therefore is most suitable to choose a p-control chart to evaluate the stability of the process.

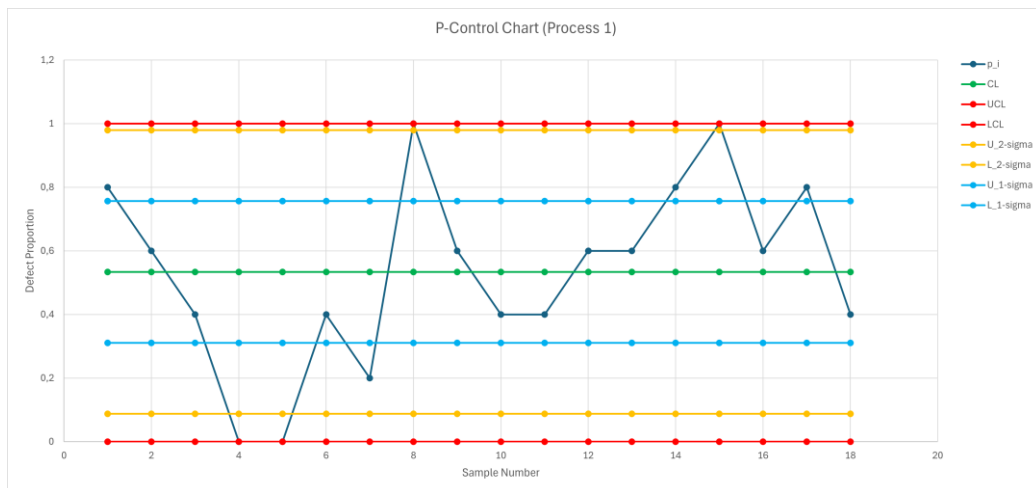


Figure 5: P-Control Chart of Process 1

For the evaluation of process stability, sensitising rules are applied according to the Western Electric rules. Points that lie on the control limits are not considered to be crossing the upper and lower control limits in this case. Evaluating the p-control chart yields the following.

Table 5: P-control Chart Process Stability Evaluation

Rule Number	Rule	Y/N
1	One or more points outside of the control limits ?	N
2	2 out of 3 consecutive points outside the two-sigma warning limits but still inside the UCL and LCL ?	N
3	4 out of 5 consecutive points beyond the one-sigma limits ?	N
4	8 consecutive points on one side of the center line ?	N

None of the rules are violated for the assessment of process stability, as shown in Table 5. The process is therefore in statistical control. All variation can be solely attributed to common causes. Therefore, the defect rate for process 1 can also be concluded to be stable at $\bar{p} = 0.53333$.

2.2 Process 2

2.2.1 Process 2: Procedure

The second process tested followed the following step to complete a tangram assembly:

1. Receive raw material sorted by colour in lin bins at the assembly station.
2. Place an assembly board retrieved from a lin bin in the centre of the assembly station.
3. Retrieve a single piece from a lin bin following a defined order. The defined order, with associated pieces indicated in Figure 1, is Red (R) → Yellow (Y) → Orange (O) → White (W) → Dark Blue (DB) → Light Green (LG) → Light Blue (LB).
4. Place the retrieved piece on the size board and evaluate if the piece fits in the shape outline (non-defective) or not (defective).
 - 4.1 If a piece is defective, it is placed on the assembly board, and the entire assembly is moved to the left of the assembly station to a demarcated area labelled "Defects". Subsequent defective assembly boards are stacked on top of each other.
 - 4.2 If a piece is non-defective, it is placed onto the assembly board and the next piece in the order is retrieved from the lin bin and evaluated for defects.
5. If no defective part is observed, the tangram is fully assembled and moved to the right of the assembly station to a demarcated area for complete assemblies.

2.2.2 Process 2: Data Collection

For the second process, the number of pieces placed onto an assembly board up until a defect is selected from the lin bin, is the recorded sample size (noted by g_i) for a specific sample. In the case of no defects being noted, seven pieces were observed as the total non-defective

pieces for a complete assembly. The defective piece's colour was also noted, and in the case of a complete assembly, "NA" would be used to indicate that no defective coloured piece was found.

Table 6: Sample Data for Process 2

Sample	Sample Size	Defective Piece Colour (g_i)
1	3	W
2	1	Y
3	1	Y
4	1	Y
5	0	R
6	0	R
7	0	R
8	0	R
9	0	R
10	0	R
11	0	R
12	1	Y
13	1	Y
14	1	Y
15	0	R
16	7	NA
17	1	Y
18	2	O
19	2	O
20	0	R
21	0	R
22	0	R
23	0	R
24	3	W
25	2	O

From the raw data, the total pieces tested was computed as:

$$\sum_{i=1}^n g_i + 1 . \quad (6)$$

Exactly one assembly run was successful in this process. Excluding that run, there is exactly

one defect per run, following from the process procedure. For process 2, the empirical defect probability ($\hat{p}_2$) can be calculated as:

$$\hat{p}_2 = \frac{\text{Total Samples}}{\text{Total pieces tested}} = \sum_{i=1}^{25} \frac{25}{g_i + 1}. \quad (7)$$

Table 7: Summary of Process 2 Data Collected

Total Samples	25
Total Pieces Tested	51
Total Defective Tangrams	24
Total Discarded Pieces (incl. defects)	43
Total Completed Tangrams	1
Defect Probability ($\hat{p}_2$)	0,4902

2.2.3 Process 2: Data Visualisation

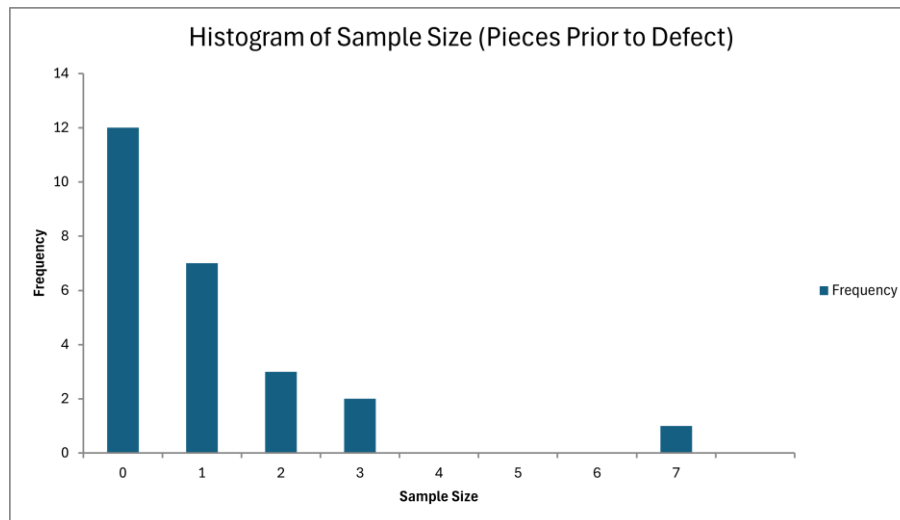


Figure 6: Histogram of Sample Size (Process 2)

The sample size recorded in Table 6 was plotted on the histogram shown in Figure 6. Following the defined order outlined in Section 2.2.1 in which the pieces were retrieved, each bin sample size correlates to a coloured piece. The only exception to this is in the case of a complete assembly as noted previously.

The highest frequency of non-defective parts placed down, noted at 12 samples, was zero. This indicates that the red piece was placed first the most frequently and was found to be defective on most of the samples. The implication of this is that the supplied red pieces have a high probability of defect.

For seven discarded assemblies, exactly one good piece was placed down before a defective piece was drawn. This frequency of yellow pieces is still relatively high in comparison to the

remainder of the pieces. The orange pieces were the result of two assemblies being discarded due to defective pieces. The white pieces were the result of the discarded assembly of a single piece. No defects were observed in the dark blue, light green, and light blue pieces.

The data presented in Figure 6 indicates data biased to the right, with a decreasing frequency shape. Early assembly failures are more common than complete assembly runs, as noted by the single recorded complete assembly with a sample size of 7.

The high frequency of red pieces being the cause of half of the discarded assemblies may require additional investigation into the supplier of the red pieces to determine the cause of these defects. The combination of red and yellow pieces that is the cause of more than half of the discarded assemblies may also require the supplier of yellow pieces to also be investigated to identify the cause of high defects.

The lack of defects in dark blue, light green and light blue pieces may imply that the suppliers of these pieces provide higher quality pieces than the suppliers of the defective pieces.

2.2.4 Process 2: Probability Distribution

For the second process, the number of non-defective pieces prior to a first defect was recorded as the sample size g_i , following the defined colour order. Assume that each draw is independent and has the same probability (p_2) of being defective. Pairing this assumption with the observation of the decaying frequency of increasing sample size, it therefore follows that process 2 follows a geometric distribution.

The probability mass function is given by:

$$P_{g_i = x} = (1 - p_2)^x p_2, \quad x = 0, 1, 2, 3, 4, 5, 6, 7 \quad (8)$$

The mean and variance of process 2 is given by:

$$E_{g_i} = \frac{1 - p_2}{p_2} \quad (9)$$

$$\text{Var } g_i = \frac{1 - p_2}{p_2^2} \quad (10)$$

The estimate of the distribution value of p_2 , $\hat{p}_2$, is calculated using equation 7 as 0.4902. Substituting $\hat{p}_2$ into equations 9 and 10 results in a sample mean value of 1.04, and a sample variance of 2.1215.

2.2.5 Process 2: Statistical Control Analysis

Process 2 follows a geometric distribution. The control chart required for this process must therefore be able to monitor the number of consecutive non-defective units prior to a defect. The measured data itself is also a measurement of an attribute as with procedure 1. A G-chart

has been selected as the most appropriate choice for establishing whether process 2 is in statistical control.

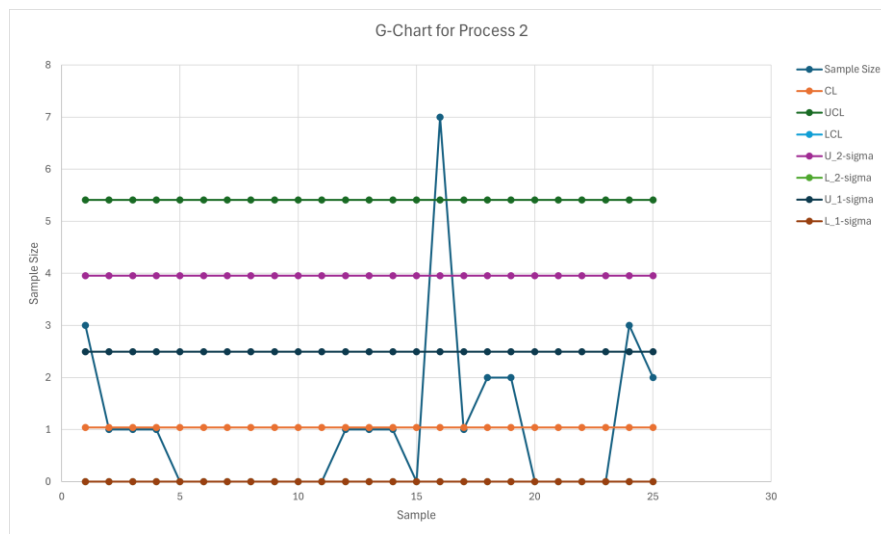


Figure 7: G-Chart of Process 2

Table 8: G-Chart Process Stability Evaluation

Rule Number	Rule	Y/N
1	One or more points outside of the control limits ?	Y
2	2 out of 3 consecutive points outside the two-sigma warning limits but still inside the UCL and LCL ?	N
3	4 out of 5 consecutive points beyond the one-sigma limits ?	N
4	8 consecutive points on one side of the center line ?	N

The single point outside of the control limits violates the sensitising rules. Points lying on the lower control limit are evaluated as not violating the control limits similar to process 1. The point outside the upper control limit can be attributed to the one successful assembly recorded. Removing the point outside the upper control limit yields the following G-chart:

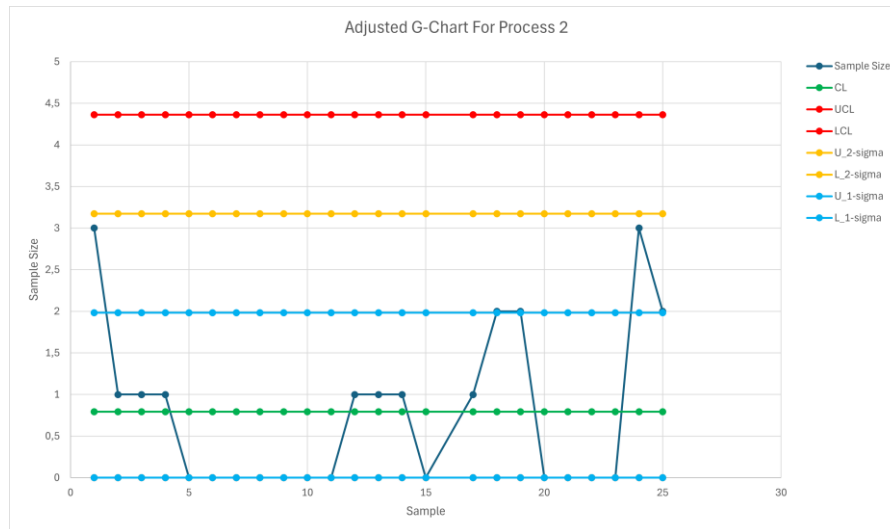


Figure 8: Adjusted G-chart of Process 2

Table 9: Adjusted G-Chart Process Stability Evaluation

Rule Number	Rule	Y/N
1	One or more points outside of the control limits ?	N
2	2 out of 3 consecutive points outside the two-sigma warning limits but still inside the UCL and LCL ?	N
3	4 out of 5 consecutive points beyond the one-sigma limits ?	N
4	8 consecutive points on one side of the center line ?	N

Removing the point outside the control limit stabilises the behaviour of the process as shown in Figure 8. The adjusted control limits, centre line, 2- σ and 1- σ lines are now slightly lower than the control chart in Figure 7. With no violations of the sensitising rules, process 2 can therefore be considered to be in statistical control with a stable defect probability p_2 over time. All variation can be attributed to common causes.

2.3 Process Capability Analysis

Both processes measure attribute data in terms of defects in each process. For a comparison on the two processes' capability, the Defects Per Million Opportunity (DPMO) of each process is compared and translated into a σ -score to compare which process performs potentially better than the other.

Table 10: Process Capability Metrics Between Process 1 and Process 2

Metric	Process 1	Process 2
$\hat{p}$ (DPU)	0,5333	0,5581
sigma level	-0,0837	-0,1462
DPMO	533333	558100
Meets 2-sigma level ?	No	No

Both processes do not meet the required 2-sigma level specification. Their negative sigma levels can be attributed to their $\hat{p}$ -values being greater than 50%. Process 1 does have a marginally lower DPMO than Process 2, with a DPMO of 533333 compared to Process 2's DPMO of 558100. In this case, if either of the two processes were to be implemented, the company would still not be able to produce enough tangrams to meet demand. However, to establish whether Process 1 actually performs better than Process 2, hypothesis testing can be performed on the sample proportions.

2.4 Hypothesis Testing

Table 11: Hypothesis Tested

Null Hypothesis	Alternative Hypothesis
$H_0: p_1 = p_2$	$H_1: p_1 > p_2$

Table 12: Two-Sample Z-Test for Proportions

Metric	Value
Pooled proportion ($\hat{p}$)	0,5414
Standard Error (SE)	0,0924
Z-Statistic (Z)	-0,2684
Significance Level (α)	0,05
p-value (One-tailed)	0,6058
Decision	Fail to Reject H_0

From the hypothesis testing conducted on the sample proportions using a two-sample Z-test for proportions with significance level of $\alpha = 0.05$, we Fail to Reject H_0 . The calculated p-value in Table 12 is shown to be significantly higher than the alpha level. There is therefore no statistical evidence to suggest that Process 1 performs significantly better than Process 2 in terms of its process capability.

2.5 Effects on Profitability

Table 13: Cost of Raw Materials, Defects, and Selling Price Information

Item	Cost (R)
Puzzle Piece	2
Assembly Board	5
Defective Semi-Assembled Item	20
Selling Price Per Completed Item (Income)	25

Table 14: Profit / Loss from Process 1

Process 1	Value	Rands (R)
Total Puzzle Pieces	90	
Total Assembly Boards	5	
Total Cost		205
Total Assembled	5	
Revenue From Sales		125
Profit / Loss		-205

Table 15: Profit / Loss from Process 2

Process 2	Value	Rands (R)
Defective Semi-Assembled Items	24	
Total Cost		480
Total Assembled	1	
Revenue From Sales		25
Profit / Loss		-480

For the financial information reported, both processes indicate that they operate at a loss to the company. Process 1 operates at a lower loss than Process 2. It is assumed that the costing for process 2 already includes the cost of assembly boards and puzzle pieces in the cost of the discarded defective semi-assembled items. If these costs were also factored in, then the reported loss of process 2 would be much greater than what it currently is compared to process 1.

Based solely on financial information, it would be best not to implement any of the two proposed processes. If one had to be implemented, process 1 would operate at a lower loss than process 2 and can thus be implemented.

2.6 Data Collection and Analysis Summary

Both processes were tested and evaluated using quality management and statistical process control techniques. In both processes, it was observed that the red and yellow pieces were the most prominent cause of defects occurring during assemblies. Light Green and Light Blue pieces were observed to have the highest quality pieces, with no defects reported in both processes.

A total of 5 completed tangrams were recorded for process 1. Process 1 was found to be binomially distributed, with an overall defect proportion of 53.33%. The batch size per sample was too small to meet the criteria for the normal approximation. Process 1 was also found to operate within statistical control, using a p-chart, without requiring the removal of any abnormal data points.

A total of 1 completed tangram was recorded for process 2. Process 2 was found to be geometrically distributed, with an overall defect proportion of 49.02%. The process was initially found to be outside statistical control, with the cause attributed to reporting of a complete assembly. With the removal of the point, process 2 was also found to operate within statistical control. The recalculated overall proportion of defects in the G-chart used for experiment 2 was found to be 55.81%

The different proportions of defects in the process were then used to calculate the DPMO for both processes and to evaluate their sigma level and the capability of the process. Neither process met the target sigma level of 2-sigma. The comparative results indicated that process 1 performs slightly better than process 2.

One-sided hypothesis testing was performed to confirm this observation. The hypothesis tests found that there is not enough statistical evidence to suggest that process 1 performs significantly better than process 2.

The financial implications of each process were also considered. Process 1 operates at a much lower cost than Process 2. However, both processes operate at a loss for the company. If one process had to be implemented, process 1 would be preferred over process 2. However, it would ideally be recommended not to implement either of the processes currently observed to run at a loss.

3 Literature Review

Statistical process control encompasses a set of statistical methods that are used to improve quality. SPC is used to help stabilise a process and reduce variability within the specific process (Montgomery, 2020). One of the prominent techniques that form part of SPC is control charts. Among the different control charts, the p-chart also forms part of the usable options (Montgomery, 2020). The p-chart is used to monitor the fraction of defective parts produced by a process (Montgomery, 2020).

The objective of control charting is mainly to identify where variability in a process is occurring that can be attributed to occurrences within a process, and not just natural variability. Sartor and Orzes (2019) as well as Montgomery (2020) confirm that a process is considered as "in control" if only natural variability occurs within the process. An out-of-control process would still have variability attributed to something occurring within a process, called systemic variability by Sartor and Orzes (2019).

A study conducted by Gitinavard et al. (2022) applied p-charts as a decision-making tool to address a joint optimisation problem of maintenance scheduling, quality control, and buffer stock planning in a machine production system. Quality control was applied to attributes commonly found in production systems. This application implies the need for a p-chart to measure attribute characteristics (Montgomery, 2020).

The study aimed to use p-control charts to detect out-of-control states and monitor the defective ratio of selected attribute characteristics. The results of the p-charts would be used to determine whether corrective maintenance or preventative maintenance would be required. The use of p-control charts, combined with sensitivity analysis, demonstrated the effects of varying key parameters such as unit costs or quality parameters. These effects would influence the optimal values of the study's optimisation problem, as well as the control limits of the p-charts and the interval to detect an out-of-control state (Gitinavard et al., 2022).

In the case of modelling count or event data, different probability distributions can be followed. If a geometric distribution provides the best fit for the count data, then an appropriate g-control chart proposed by Kaminsky et al. (1992) can be used as an alternative to a c-chart (Montgomery, 2020).

G-charts are used mainly in monitoring rare events (Schrem et al., 2016). Sources on applications of g-charts in production environments were hard to come by for this review of the literature. The application of g-charts in environments used to monitor rare events sees it appearing in other fields more readily, such as in medical fields.

A study by Schrem et al. (2016) applied g-charts to analyse the time between consecutive cancer diagnoses. The authors of the study suggested that the use of a g-chart with alarm signals can trigger investigations into changes in cancer detection rates. Furthermore, the use of g-chart alarms could also drive processes based on identified cancer risk factors to adapt when investigations are triggered (Schrem et al., 2016).

The usefulness of comparative analysis of g-charts is also outlined in the study. The comparative analysis of g-charts is noted by Schrem et al. (2016) as being able to facilitate the benchmarking of cancer surveillance processes between different cancer centres. The usefulness of g-charts in environments where monitoring for the occurrence of rare events is well demonstrated for this study.

4 Conclusion

Both processes used for this experiment performed well below the required 2-sigma specification limit for their assembly line to meet demand. Process 1 performed better than process 2, although on a marginal level. Hypothesis testing confirmed that neither process performed significantly better than the other. Process 1 performed at a lower overall loss to the company than process 2. If one process of the two had to be selected, process 1 would be selected for its lower loss to the company and the higher number of completed tangrams produced in the same 15-minute time limit as process 2, as well as its lower reported DPMO.

Both processes were able to identify that the most defects were caused by red and yellow puzzle pieces, and the highest quality pieces were light blue and light green pieces. More research and investigation are required about the suppliers of red and yellow pieces and the cause of the large amounts of defective pieces. In addition, management could consider looking at suppliers of higher-quality pieces and adopt their strategies to address issues with suppliers of lower-quality pieces. Addressing these high-defect piece issues may improve the DPMO of the process and subsequent sigma-level closer towards the 2-sigma level.

More data is required to definitively identify which process will benefit the company the most. Larger batch sizes may be taken for process 1 to make a normal approximation possible. Process 2 may need to be re-evaluated for use when processes are in statistical control at lower defect proportions when defects occur less frequently. Evaluating each process over a longer time period to allow more observations to be collected may also provide more opportunities for better insights into the two processes tested to be obtained.

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A Raw Data

A.1 Process 1

Process 1: n=5

Sample Number,	n	# defects	Colors
1	5	4	R
2	5	3	Y
3	5	2	O.
4	5	0	LB
5	5	0.	CG
6	5	2	DB
7	5	1	W
8	5	5	R
9	5	3	R
10	5	2	Y
11	5	2	R.
12	5	3	O
13	5	3	DB
14	5	4	W
15	5	5	Y
16	5	3	Y
17	5	4	O.
18	5	2	W
19	5	Total Defects: 46.	
20	5		
21	5	Total Assembled Transponders: 5	
22	5		
23	5		
24	5		
25	5		
26	5		
27	5		
28	5		
29	5		

Figure 9: Process 1 Raw Data

A.2 Process 2

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Assembly Run	Assembly Attempt	Tm; time to Defect	# pieces to 1st defect	Colour
1	1		4	W
2	2		2	Y
	1		2	Y
	1		2	Y
	1		1	R
	1		1	R
	1		1	R
	1		1	R
	1		1	R
	1		2	Y
	1		2	Y
	1		2	Y
	1		1	R
	1		0	
2	1		2	Y
	1		2	O
	1		3	O
	1		1	R
	1		1	R
	1		1	R
	1		1	R
	1		4	W
	1		3	O
Total Defects: 24				
Total Completed: 1				

Figure 10: Process 2 Raw Data

B Control Charts

B.1 P-control Chart Control Limit and Warning Level Values

Table 16: P-Control Chart Data

Sample Number	n	Number of Defects (d _i)	Colour	p _i	UCL _{p_chart}	LCL _{p_chart}	Centre Line _{p_chart}	U _{2-sigma}	L _{2-sigma}	U _{1-sigma}	L _{1-sigma}
1	5	4	R	0,8	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
2	5	3	Y	0,6	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
3	5	2	O	0,4	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
4	5	0	LB	0	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
5	5	0	LG	0	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
6	5	2	DB	0,4	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
7	5	1	W	0,2	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
8	5	5	R	1	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
9	5	3	R	0,6	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
10	5	2	Y	0,4	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
11	5	2	R	0,4	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
12	5	3	O	0,6	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
13	5	3	DB	0,6	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
14	5	4	W	0,8	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
15	5	5	Y	1	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
16	5	3	Y	0,6	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
17	5	4	O	0,8	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022
18	5	2	W	0,4	1	0	0,533333333	0,979552	0,08711	0,756443	0,31022

B.2 G-Control Chart Data

B.2.1 G-Control Chart Data for G-chart Including Successful Run

Table 17: G-Control Chart Data before Adjustment

Sample	Sample Size	Defective Piece Colour	CL	UCL	LCL	U_2-sigma	L_2-sigma	U_1-sigma	L_1-sigma
1	3	W	1,04	5,40971	0	3,953143	0	2,4965713	0
2	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
3	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
4	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
5	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
6	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
7	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
8	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
9	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
10	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
11	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
12	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
13	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
14	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
15	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
16	7	NA	1,04	5,40971	0	3,953143	0	2,4965713	0
17	1	Y	1,04	5,40971	0	3,953143	0	2,4965713	0
18	2	O	1,04	5,40971	0	3,953143	0	2,4965713	0
19	2	O	1,04	5,40971	0	3,953143	0	2,4965713	0
20	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
21	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
22	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
23	0	R	1,04	5,40971	0	3,953143	0	2,4965713	0
24	3	W	1,04	5,40971	0	3,953143	0	2,4965713	0
25	2	O	1,04	5,40971	0	3,953143	0	2,4965713	0

B.2.2 G-Control Chart Data for G-chart Excluding Successful Run

Table 18: Adjusted G-Control Chart Data

Sample	Sample Size	Defective Piece Colour	CL	UCL	LCL	U_2-sigma	L_2-sigma	U_1-sigma	L_1-sigma
1	3	W	0,79167	4,36457	0	3,173601	0	1,9826338	0
2	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
3	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
4	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
5	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
6	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
7	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
8	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
9	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
10	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
11	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
12	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
13	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
14	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
15	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
17	1	Y	0,79167	4,36457	0	3,173601	0	1,9826338	0
18	2	O	0,79167	4,36457	0	3,173601	0	1,9826338	0
19	2	O	0,79167	4,36457	0	3,173601	0	1,9826338	0
20	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
21	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
22	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
23	0	R	0,79167	4,36457	0	3,173601	0	1,9826338	0
24	3	W	0,79167	4,36457	0	3,173601	0	1,9826338	0
25	2	O	0,79167	4,36457	0	3,173601	0	1,9826338	0